

BinX TECH

# ARTIFICIAL INTELLIGENCE & MACHINE LEARNING INTERNSHIP PROGRAM

400-Hour Internship Outline

*A hands-on program taking interns from Python fundamentals to building, evaluating, and deploying real AI/ML models — directly aligned with one of the fastest-growing fields in the global labour market.*

400 HOURS • 4 PHASES • REAL DATASETS & MODELS • DEPLOYED AI PRODUCTS

Prepared by BinX Tech  
Palestine | Nablus



## 1. Program Overview

BinX Tech's Artificial Intelligence & Machine Learning Internship is a structured, 400-hour program that takes participants from Python and mathematics fundamentals through to building, evaluating, and deploying production-ready AI and ML models. The program combines guided technical training with real dataset projects under the supervision of experienced practitioners.

The internship is built for computer science, mathematics, and engineering students, recent graduates, and early-career developers who want to transition into one of the highest-demand fields in today's technology market.

|                         |                                                                      |
|-------------------------|----------------------------------------------------------------------|
| <b>Total Duration</b>   | 400 hours                                                            |
| <b>Recommended Span</b> | 10 weeks full-time (40 hrs/week) or 20 weeks part-time (20 hrs/week) |
| <b>Format</b>           | On-site / Remote / Hybrid (configurable)                             |
| <b>Track</b>            | Artificial Intelligence & Machine Learning                           |

## 2. Objectives

### Technical Competencies

- Build a strong command of Python and the core scientific libraries used across all AI/ML work.
- Understand the mathematical foundations of machine learning: linear algebra, probability, and statistics.
- Implement, train, evaluate, and tune both classical ML and deep learning models from scratch and using industry-standard frameworks.
- Work through the full ML pipeline: data collection, cleaning, EDA, feature engineering, modelling, evaluation, and deployment.

### Professional & Soft Skills

- Develop strong analytical thinking and the ability to form and test hypotheses against real data.
- Communicate model results and technical findings clearly to both technical and non-technical audiences.
- Build habits of experiment documentation, reproducibility, and structured problem-solving.

### Market-Readiness Outcomes

- Leave with a portfolio of real, deployed AI/ML projects demonstrating end-to-end ownership.
- Acquire skills directly aligned with current AI/ML job postings: Python, Scikit-learn, TensorFlow/PyTorch, SQL, and deployment tools.
- Graduate with the confidence and credentials to compete for ML Engineer, Data Scientist, or AI Developer roles.



### 3. Program Structure (Phases)

The 400 hours are divided into four progressive phases. Each phase builds on the last, moving the intern from Python and math fundamentals to independent, real-world AI/ML model delivery.

| Phase                                     | Hours          | Focus                                                            |
|-------------------------------------------|----------------|------------------------------------------------------------------|
| Phase 1 — Foundations                     | 50 hrs         | Python, math refresher, data manipulation & EDA                  |
| Phase 2 — Core ML Training                | 140 hrs        | Classical ML algorithms, feature engineering, model evaluation   |
| Phase 3 — Deep Learning & Applied Project | 170 hrs        | Neural networks, DL frameworks, real project sprints, deployment |
| Phase 4 — Final Evaluation & Wrap-up      | 40 hrs         | Final presentation, portfolio review, certification              |
| <b>Total</b>                              | <b>400 hrs</b> |                                                                  |

#### Weekly Hour Distribution & Timeline

The standard track runs full-time across 10 weeks at 40 hours/week (8 hrs/day, 5 days). A part-time track runs 20 weeks at 20 hrs/week, following the same phase sequence and proportions.

| Week | Hours  | Phase       | Focus                                                                      |
|------|--------|-------------|----------------------------------------------------------------------------|
| 1    | 40 hrs | Phase 1     | Python environment, NumPy, Pandas, Matplotlib, Jupyter                     |
| 2    | 40 hrs | Phase 1 → 2 | Statistics, probability, linear algebra refresher; EDA on real datasets    |
| 3    | 40 hrs | Phase 2     | Supervised learning: regression & classification (Scikit-learn)            |
| 4    | 40 hrs | Phase 2     | Model evaluation, cross-validation, feature engineering & pipelines        |
| 5    | 40 hrs | Phase 2 → 3 | Unsupervised learning, clustering, dimensionality reduction; begin project |
| 6    | 40 hrs | Phase 3     | Neural networks & deep learning intro (TensorFlow/PyTorch); Sprint 1       |
| 7    | 40 hrs | Phase 3     | CNNs / RNNs / Transformers intro; Sprint 2 — core model development        |



|              |        |                |                                                                         |
|--------------|--------|----------------|-------------------------------------------------------------------------|
| 8            | 40 hrs | Phase 3        | NLP & Computer Vision fundamentals; Sprint 3 — integration & evaluation |
| 9            | 40 hrs | Phase 3        | Model deployment (FastAPI + Streamlit); Sprint 4 — polish & deploy      |
| 10           | 40 hrs | Phase 4        | Final presentation, portfolio review, feedback & certification          |
| <b>Total</b> |        | <b>400 hrs</b> |                                                                         |

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## 4. Phase 3 Spotlight — Real Project Examples

During Phase 3, each intern (or small team) builds one complete AI/ML project end-to-end — from raw data to a deployed, usable product. Mentors assign one of the following project types, or a comparable live BinX Tech use case, based on intern interest and current team needs:

- Customer Churn Prediction System — binary classification model that predicts whether a customer will leave a service, using real telecom or SaaS datasets. Includes EDA, feature importance, and threshold tuning.
- House Price Prediction — regression model trained on real estate data (e.g., Ames Housing dataset). Covers preprocessing pipelines, polynomial features, and regularization techniques.
- Sentiment Analysis Tool — NLP model that classifies the sentiment of product reviews, tweets, or support tickets (positive / neutral / negative). Uses TF-IDF or a pre-trained transformer from Hugging Face.
- Image Classifier — CNN-based model trained to classify images into categories (e.g., plant disease detection, medical image classification, or handwritten digit recognition). Demonstrates data augmentation and transfer learning.
- Recommendation System — collaborative or content-based filtering model that suggests products, movies, or content to users based on historical interaction data.
- Fraud Detection Model — imbalanced classification task on financial transaction data. Covers SMOTE oversampling, precision-recall trade-offs, and model explanation using SHAP values.

**Every project, regardless of type, must meet the following professional baseline:**

- A clean, well-documented Jupyter Notebook covering the full ML pipeline: EDA → preprocessing → modelling → evaluation.
- A trained and evaluated model with reported metrics (accuracy, F1-score, AUC-ROC, RMSE, or equivalent, depending on task type).
- A working deployment: a Streamlit dashboard or FastAPI endpoint accessible via a public URL (hosted on Hugging Face Spaces, Render, or Railway).
- A GitHub repository with a clean README, requirements.txt, model artifacts, and experiment logs.
- A short technical write-up explaining approach, model choices, evaluation results, and limitations.

## 5. Weekly / Module Breakdown

Topics are sequenced to mirror how a working ML engineer or data scientist ramps up in a real team, with each module chosen for its direct relevance to current AI/ML job-market demand.

| Module                  | Key Topics                                                                                            |
|-------------------------|-------------------------------------------------------------------------------------------------------|
| Python for Data Science | Python ES6-equivalent syntax, list comprehensions, functions, OOP; Jupyter Notebook workflow          |
| Data Manipulation       | NumPy arrays, Pandas DataFrames, data cleaning, merging, grouping, and reshaping                      |
| Math Foundations        | Linear algebra (vectors, matrices, dot products), descriptive statistics, probability & distributions |

|                                 |                                                                                                                  |
|---------------------------------|------------------------------------------------------------------------------------------------------------------|
| Exploratory Data Analysis (EDA) | Matplotlib, Seaborn, correlation analysis, outlier detection, data storytelling                                  |
| Supervised Learning             | Linear & logistic regression, decision trees, random forests, SVMs, k-NN (Scikit-learn)                          |
| Model Evaluation & Tuning       | Train/val/test splits, cross-validation, confusion matrix, F1, AUC-ROC, GridSearchCV, pipelines                  |
| Unsupervised Learning           | K-Means, DBSCAN, PCA, t-SNE, hierarchical clustering, anomaly detection                                          |
| Deep Learning Fundamentals      | Neural network architecture, backpropagation, activation functions, optimizers, batch normalization              |
| CNNs, RNNs & Transformers       | Convolutional networks for images, recurrent networks for sequences, attention mechanism, pre-trained models     |
| NLP Fundamentals                | Tokenization, TF-IDF, word embeddings (Word2Vec, GloVe), Hugging Face Transformers for text classification       |
| Computer Vision Basics          | Image preprocessing with OpenCV, data augmentation, transfer learning with ResNet/EfficientNet                   |
| Model Deployment                | FastAPI for model serving, Streamlit for interactive dashboards, hosting on Hugging Face Spaces or Railway       |
| MLOps & Experiment Tracking     | MLflow for experiment logging, model versioning, reproducibility best practices, requirements.txt & environments |



## 6. Technical Stack & Tools

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A detailed breakdown of every language, library, framework, and tool interns will work with hands-on throughout the program.

### Programming Language

- Python 3.10+ — the primary and industry-standard language for AI/ML. Interns will use Pythonic coding patterns: list comprehensions, generators, decorators, type hints, and virtual environments.

### Data Manipulation & Analysis

- NumPy — array operations, linear algebra, broadcasting, and vectorized computation.
- Pandas — DataFrames, data cleaning, merging, groupby aggregations, and time-series handling.
- SQL (SQLite / MySQL) — data querying for structured datasets: SELECT, JOIN, GROUP BY, subqueries, and window functions.

### Data Visualization

- Matplotlib — core plotting library: line, scatter, bar, histogram, and subplots.
- Seaborn — statistical visualization: heatmaps, pairplots, violin plots, and distribution charts.
- Plotly — interactive dashboards and charts inside Streamlit apps.

### Classical Machine Learning

- Scikit-learn — the complete ML toolkit: estimators, transformers, pipelines, cross-validation, grid search, and evaluation metrics.
- Algorithms covered: Linear Regression, Logistic Regression, Decision Trees, Random Forests, Gradient Boosting (XGBoost/LightGBM), SVM, k-NN, k-Means, PCA, DBSCAN.
- SHAP — model explainability and feature importance visualization.

### Deep Learning Frameworks

- TensorFlow / Keras — building, training, and evaluating neural networks using the Sequential and Functional API; callbacks, early stopping, and model checkpointing.
- PyTorch — tensor operations, autograd, custom Dataset/DataLoader classes, training loops, and model serialization.
- Hugging Face Transformers — loading and fine-tuning pre-trained NLP models (BERT, DistilBERT, GPT-2) for classification and generation tasks.

### Computer Vision

- OpenCV — image reading, resizing, color conversion, edge detection, and basic augmentation.
- Torchvision / tf.keras.preprocessing — image data generators, data augmentation pipelines, and transfer learning with ResNet, EfficientNet, and MobileNet.

### Model Deployment & APIs

- FastAPI — building REST API endpoints that serve trained model predictions; request/response handling, Pydantic validation, and async support.

- Streamlit — building interactive AI-powered web apps and dashboards directly from Python scripts for project demos.
- Hugging Face Spaces / Railway — hosting and sharing deployed ML applications publicly with automatic builds.

### Experiment Tracking & Reproducibility

- MLflow — logging experiments, parameters, metrics, and model artifacts; comparing runs across experiments.
- requirements.txt / conda environments — dependency management and environment reproducibility.
- Random seeds and deterministic training — ensuring experiment repeatability across runs.

### Development Environment & Workflow

- Jupyter Notebook / JupyterLab — the primary interactive development environment for experimentation and documentation.
- Google Colab — free GPU/TPU access for training deep learning models; integration with Google Drive.
- Visual Studio Code (VS Code) — IDE for production-grade Python scripts; extensions: Python, Pylance, Jupyter, GitLens.
- Git & GitHub — version control, branching strategy (feature branches + pull requests), structured code review, and GitHub Actions basics for CI.
- Command line / terminal — virtual environments (venv/conda), package management (pip), and script execution.

### Version Control

- Git — init, clone, add, commit, branch, merge, and conflict resolution.
- GitHub — remote repositories, pull requests, Issues, GitHub Projects (Kanban board), and code review.
- Branching strategy: feature branches per experiment/sprint, merged to main via reviewed pull requests.

## 7. Sprint Criteria & Definition of Done

Phase 3 runs across 4 one-week sprints. Each sprint follows the same structured cycle used on professional AI/ML teams, with clear entry criteria, a Definition of Done, and a formal review before the next sprint begins.

### Sprint Structure (Per Sprint)

| Event                                      | Description                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sprint Planning (Day 1)                    | Intern selects tasks from the ML project backlog (EDA, modelling, evaluation, deployment), estimates effort, and commits to a sprint goal.                   |
| Daily Stand-up (Daily)                     | 3-minute update: what was completed, what is next, any blockers (data issues, model errors, environment problems).                                           |
| Mentor Code & Notebook Review (Mid-sprint) | Mentor reviews the Jupyter Notebook and code via GitHub pull request; provides structured comments on methodology, code quality, and results interpretation. |
| Sprint Review (Day 5)                      | Intern demos completed pipeline stages or model results to the mentor. Incomplete tasks are moved to the next sprint backlog with documented reasons.        |
| Sprint Retrospective (Day 5)               | What went well, what to improve, and one specific action for the next sprint (e.g., try a different algorithm, improve preprocessing).                       |

### Acceptance Criteria (Per Task)

Every task on the sprint board must have written acceptance criteria before work begins. A task is only considered done when ALL of the following are true:

- The notebook cell or script produces the expected output without errors.
- Code is committed to the correct feature branch with a clear, descriptive commit message referencing the task.
- A pull request has been opened and approved by the mentor before merging.
- Results are documented in the notebook with clear markdown cells explaining findings, choices, and next steps.
- Model metrics are logged (in MLflow or directly in the notebook) and compared to the baseline.

### Definition of Done (Full Project)

The AI/ML project is considered complete and eligible for final presentation when ALL of the following are met:

- All sprint tasks are marked complete on the board and the full ML pipeline is implemented end-to-end.
- The model achieves a documented, reproducible performance benchmark (e.g.,  $F1 \geq 0.80$ , RMSE within acceptable range for the dataset).



- The model is deployed: accessible via a live Streamlit app or FastAPI endpoint at a public URL.
- The GitHub repository has a clean README (problem statement, dataset, methodology, results, setup instructions) and a requirements.txt.
- All experiments are logged in MLflow or documented in the notebook with parameters, metrics, and model version.
- The notebook is clean, well-commented, and cell outputs are present and reproducible with a fixed random seed.

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## 8. Attendance & Absence Policy

Consistent attendance is a core professional requirement of this program. The minimum attendance threshold to remain eligible for certification is 90% of total program hours (360 of 400 hours).

### Absence Tiers & Consequences

| Absence Count  | Action                                                                                                                                                                 |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 – 2 absences | Logged. No formal action; intern is expected to make up missed hours within the same week.                                                                             |
| 3 absences     | First Written Warning — issued by mentor. A formal note is added to the intern's record and a corrective conversation is held.                                         |
| 5 absences     | Final Warning — mandatory meeting with mentor and program coordinator. A written improvement plan is agreed upon. Continued absence will result in dismissal.          |
| 7+ absences    | Dismissal from Program — the intern is withdrawn from the cohort. No certificate is issued. The intern may reapply for a future cohort after a 3-month waiting period. |

### Late Arrivals

- An arrival of more than 15 minutes late without prior notice is recorded as a late arrival.
- 3 late arrivals = 1 recorded absence and count toward the absence tier above.
- Arriving more than 2 hours late without notice is recorded directly as a full absence.

### Excused Absences

- Absences due to illness, documented emergency, or pre-approved personal circumstances are considered excused.
- Excused absences must be communicated to the mentor at least 2 hours before the session begins (except in emergencies).
- Excused absences still count toward the 90% threshold but are excluded from the warning tier counts.
- Supporting documentation (e.g., medical certificate) may be requested for multiple excused absences in the same week.

### Makeup Hours Policy

- Any missed hours (excused or unexcused) may be made up within 7 days, subject to mentor approval.
- Makeup sessions must be agreed in advance and logged on the attendance record.
- Makeup hours cannot be used to exceed 40 hours in a single week.

## 9. Mentorship & Supervision



- Each intern is assigned a dedicated ML mentor for the full 400 hours.
- Weekly one-on-one check-ins to review model progress, discuss blockers, and set goals.
- Structured notebook and code reviews on all submitted work, following real AI/ML team standards.
- Direct access to senior practitioners for guidance on methodology, model selection, and deployment.

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## 10. Skills Matrix

| Technical Skills (Market-Aligned)                | Professional Skills                               |
|--------------------------------------------------|---------------------------------------------------|
| Python & scientific libraries (NumPy, Pandas)    | Analytical thinking & hypothesis testing          |
| Data cleaning, EDA & visualization               | Communicating findings to non-technical audiences |
| Classical ML algorithms (Scikit-learn)           | Experiment documentation & reproducibility        |
| Deep learning (TensorFlow / PyTorch)             | Problem-solving with ambiguous, real-world data   |
| NLP & Computer Vision fundamentals               | Time management & ownership under deadlines       |
| Model evaluation, tuning & explainability (SHAP) | Receiving & applying structured feedback          |
| Model deployment (FastAPI, Streamlit, Railway)   | Presenting technical results with clarity         |
| MLflow experiment tracking & Git/GitHub          | Working within a professional team structure      |
| SQL for data querying & Agile/Scrum workflow     | Writing technical documentation & reports         |

## 11. Deliverables

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- Mini-projects at the end of each core module (EDA notebook, classification model, regression model, clustering analysis).
- One complete AI/ML capstone project built end-to-end during Phase 3, across 4 sprints.
- A clean, documented GitHub repository with the full project (notebooks, scripts, README, requirements.txt, model artifacts).
- A deployed, live ML application accessible via a public URL (Streamlit or FastAPI).
- MLflow experiment log documenting all training runs, parameters, and metrics comparisons.
- A written technical report: problem statement, dataset description, methodology, results, limitations, and future work.
- A final presentation of the capstone project, including a live demo, to BinX Tech staff.

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## 12. Evaluation & Internal Scoring System

The evaluation system has three layers: (1) overall criterion weights, (2) a detailed scoring rubric on a 100-point scale, and (3) a week-by-week tracker that produces the intern's final grade.

### Evaluation Weights

| Criteria                           | Weight | Assessed By           |
|------------------------------------|--------|-----------------------|
| Technical skill development        | 40%    | Mentor                |
| Sprint / project quality & results | 30%    | Mentor + Review panel |
| Professionalism & consistency      | 20%    | Mentor                |
| Final presentation (Week 10)       | 10%    | Review panel          |



## Internal Scoring Rubric — 100-Point Scale

Each criterion is scored out of 100 by the assigned evaluator each week. The final weekly score is the weighted average of all active criteria for that week. Criteria marked 'N/A' are excluded from that week's calculation.

| Criterion                                | 0 – 49 Below Standard                 | 50 – 69 Developing                    | 70 – 84 Proficient               | 85 – 100 Excellent                                       |
|------------------------------------------|---------------------------------------|---------------------------------------|----------------------------------|----------------------------------------------------------|
| Python code quality & notebook structure | Messy code, no markdown documentation | Basic structure, some comments        | Clean, readable, well-documented | Production-grade, fully reproducible                     |
| Understanding of ML concepts             | Cannot explain model choices          | Partial understanding of algorithms   | Explains most choices clearly    | Deep understanding, can justify trade-offs               |
| Data preprocessing & EDA quality         | Raw data used without cleaning        | Basic cleaning, shallow EDA           | Thorough EDA with clear insights | Comprehensive EDA, feature engineering, insight-driven   |
| Model evaluation & metric reporting      | No evaluation or wrong metrics        | Single metric, no baseline comparison | Multiple metrics with baseline   | Full evaluation + confusion matrix + SHAP/explainability |
| Model completeness & performance         | Model does not train or predict       | Trains but below baseline             | Meets benchmark, properly tuned  | Exceeds benchmark, optimized & justified                 |
| Git workflow & experiment tracking       | No commits or experiments logged      | Sporadic commits, no tracking         | Regular commits + MLflow logs    | Consistent branching, full experiment history            |
| Deployment (Streamlit / FastAPI)         | Not deployed                          | Partial or local-only deployment      | Live public URL, functional      | Polished UI, fast, handles edge cases                    |
| Attendance & punctuality                 | 7+ absences / chronic lateness        | 3–6 absences                          | 1–2 absences, on time            | Perfect attendance, proactive                            |
| Responsiveness & initiative              | Waits to be told what to do           | Responds when prompted                | Self-directed most of the time   | Proactively identifies & solves problems                 |
| Final presentation quality               | Cannot explain the project            | Partial explanation, demo breaks      | Clear explanation, working demo  | Confident, polished, handles Q&A                         |



## Weekly Evaluation Tracker — 10 Weeks

The mentor scores each active criterion (out of 100) at the end of every week. Final grade: (average of weeks 1–9) × 90% + (week 10 presentation) × 10%.

| Week               | Phase & Focus                                               | Technical /100 | Professional. /100 | Sprint/Deliverable /100 | Weekly Avg /100 |
|--------------------|-------------------------------------------------------------|----------------|--------------------|-------------------------|-----------------|
| 1                  | Phase 1 — Python, NumPy, Pandas, Jupyter                    | —              | —                  | N/A                     | —               |
| 2                  | Phase 1→2 — Math & EDA on real datasets                     | —              | —                  | N/A                     | —               |
| 3                  | Phase 2 — Supervised learning (regression & classification) | —              | —                  | Mini-project 1          | —               |
| 4                  | Phase 2 — Model evaluation, tuning & pipelines              | —              | —                  | Mini-project 2          | —               |
| 5                  | Phase 2→3 — Unsupervised learning + begin project           | —              | —                  | Sprint 1 planning       | —               |
| 6                  | Phase 3 — Deep learning intro; Sprint 1 review              | —              | —                  | Sprint 1 /100           | —               |
| 7                  | Phase 3 — CNNs / Transformers; Sprint 2 review              | —              | —                  | Sprint 2 /100           | —               |
| 8                  | Phase 3 — NLP / CV; Sprint 3 review                         | —              | —                  | Sprint 3 /100           | —               |
| 9                  | Phase 3 — Deploy + polish; Sprint 4 review                  | —              | —                  | Sprint 4 /100           | —               |
| 10                 | Phase 4 — Final presentation & certification                | N/A            | N/A                | Presentation /100       | —               |
| <b>FINAL GRADE</b> | <b>(Wks 1–9 avg × 90%) + (Wk 10 × 10%)</b>                  |                |                    |                         | <b>— / 100</b>  |



## Passing Grade & Outcomes

A minimum final grade of 70/100 is required to pass and receive the BinX Tech Certificate of Completion.

| Final Grade | Outcome                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|
| 85 – 100    | Outstanding — Certificate of Completion + Letter of Recommendation + priority consideration for AI/ML roles at BinX Tech.     |
| 70 – 84     | Pass — Certificate of Completion awarded.                                                                                     |
| 50 – 69     | Conditional Pass — structured improvement plan issued by mentor; certificate awarded once all gaps are formally closed.       |
| Below 50    | Does Not Pass — no certificate issued; detailed written feedback provided. Case reviewed for one of the three outcomes below. |

## Program Status Review — Failing & Special Cases

When an intern falls below the passing threshold — whether due to academic performance, attendance, or conduct — the program coordinator convenes a formal review within 5 business days. The committee issues one of three decisions:

| Decision                              | Description & Conditions                                                                                                                                                                                                        |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Continue with Remediation Plan</b> | The intern continues under a formal, time-bound improvement plan with weekly milestone targets. Applicable when the intern shows genuine effort and the gap is recoverable within the remaining program time.                   |
| <b>Program Repeat</b>                 | The intern is withdrawn from the current cohort but formally invited to re-enroll in the next cohort, starting from the beginning. A 3-month waiting period applies.                                                            |
| <b>Permanent Dismissal</b>            | The intern is permanently removed from the BinX Tech Internship Program with no right to re-apply. Applied in cases of repeated violations, academic dishonesty, or a second failed attempt. A formal written notice is issued. |



## 13. Career & Labour Market Impact

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**AI and Machine Learning is the fastest-growing domain in the global technology market. Every element of this program is designed to translate directly into stronger job-market positioning for the intern.**

- Official BinX Tech Certificate of Completion in AI/ML — a credible, verifiable credential for employers.
- Letter of recommendation for top-performing interns.
- A deployed, live AI/ML project portfolio that interns can demonstrate in interviews, on GitHub, and on LinkedIn.
- Curriculum built around current, in-demand skills: Python, Scikit-learn, TensorFlow/PyTorch, Hugging Face, FastAPI, Streamlit — not outdated theory.
- Direct hands-on experience with the tools and workflows used on real AI/ML teams: Jupyter, Git, MLflow, Agile sprints, code review.
- CV and LinkedIn profile guidance based on projects and skills completed during the program.
- Industry exposure and networking with BinX Tech's AI/ML team.
- A possible pathway to paid ML Engineering, Data Science, or AI Development roles at BinX Tech for outstanding interns.

## 14. Eligibility & Requirements

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### Prerequisites

- Currently enrolled in or graduated from a Computer Science, Mathematics, Statistics, or Engineering program (or equivalent self-taught background).
- Basic familiarity with programming in any language (Python experience preferred but not required).
- Foundational understanding of high-school level mathematics (algebra and basic statistics).
- A laptop capable of running Python, Jupyter Notebook, and a standard development environment.
- Strong curiosity, analytical mindset, and willingness to work with data, uncertainty, and iteration.

### Application Process

- Submit application & CV.
- Short technical screening (Python basics + a simple data manipulation task).
- Interview with the BinX Tech AI/ML team.
- Offer & onboarding into the 400-hour program.